Multiple comparisons

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete comparison-family and decision record.

Keep family size, supplied p-values, rule, missing results and inferential limits together.

Keep one habit: Recall → W1–W3/W5 → new four-test family → research project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

Statistical tests and assumptions

> **My remembered idea, example and correction**

Uncertainty intervals

> **My remembered idea, example and correction**

Experimental units and independence

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Bonferroni controls familywise error: the chance of at least one false rejection among true nulls is at most family alpha when individual p-values are valid. Tests need not be independent for the union bound. Declare the complete family and compare each p with alpha divided by the number of tests, including equality. Main five p-values 0.009, 0.012, 0.3, 0.4, 0.8 use cutoff 0.01; only 0.009 rejects.

  Separate picture p-values 0.004, 0.012, 0.02, 0.08, 0.4, 0.9 use six-test cutoff 1/120, approximately 0.00833; only 0.004 rejects. Changed ten-test family has cutoff 0.005, with four outcomes missing; only 0.004 among supplied values rejects, and the other four decisions are unknown. Keep those tests in the denominator and disclose missing results. The scenarios are stipulated, not actual experiments.

  An adjusted p-value is min(1, m × p). Adjustment cannot validate invalid p-values, hide outcome-driven selection, prove non-rejected nulls or establish useful effect sizes. The error bound can be conservative and is not an observed error rate.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- Bonferroni controls familywise error: the chance of at least one false rejection among true nulls is at most family alpha when individual p-values are valid. Tests need not be independent for the union bound. Declare the complete family and compare each p with alpha divided by the number of tests, including equality. Main five p-values 0.009, 0.012, 0.3, 0.4, 0.8 use cutoff 0.01; only 0.009 rejects.

  Separate picture p-values 0.004, 0.012, 0.02, 0.08, 0.4, 0.9 use six-test cutoff 1/120, approximately 0.00833; only 0.004 rejects. Changed ten-test family has cutoff 0.005, with four outcomes missing; only 0.004 among supplied values rejects, and the other four decisions are unknown. Keep those tests in the denominator and disclose missing results. The scenarios are stipulated, not actual experiments.

  An adjusted p-value is min(1, m × p). Adjustment cannot validate invalid p-values, hide outcome-driven selection, prove non-rejected nulls or establish useful effect sizes. The error bound can be conservative and is not an observed error rate.

- Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality. Show the intermediate steps and give the conclusion with its scope.

Main test	Supplied p	Bonferroni cutoff	Decision
H1	0.009	0.010	Reject
H2	0.012	0.010	Do not reject
H3	0.300	0.010	Do not reject
H4	0.400	0.010	Do not reject
H5	0.800	0.010	Do not reject

Main test	Supplied p	Bonferroni cutoff	Decision
Unadjusted comparison	H1 and H2 are ≤ 0.05	This is a different decision rule	Does not provide the same familywise bound

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Bonferroni controls familywise error: the chance of at least one false rejection among true nulls is at most family alpha when individual p-values are valid. Tests need not be independent for the union bound. Declare the complete family and compare each p with alpha divided by the number of tests, including equality. Main five p-values 0.009, 0.012, 0.3, 0.4, 0.8 use cutoff 0.01; only 0.009 rejects.

  Separate picture p-values 0.004, 0.012, 0.02, 0.08, 0.4, 0.9 use six-test cutoff 1/120, approximately 0.00833; only 0.004 rejects. Changed ten-test family has cutoff 0.005, with four outcomes missing; only 0.004 among supplied values rejects, and the other four decisions are unknown. Keep those tests in the denominator and disclose missing results. The scenarios are stipulated, not actual experiments.

  An adjusted p-value is min(1, m × p). Adjustment cannot validate invalid p-values, hide outcome-driven selection, prove non-rejected nulls or establish useful effect sizes. The error bound can be conservative and is not an observed error rate.

- A report omits this qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Bonferroni controls familywise error: the chance of at least one false rejection among true nulls is at most family alpha when individual p-values are valid. Tests need not be independent for the union bound. Declare the complete family and compare each p with alpha divided by the number of tests, including equality. Main five p-values 0.009, 0.012, 0.3, 0.4, 0.8 use cutoff 0.01; only 0.009 rejects.

  Separate picture p-values 0.004, 0.012, 0.02, 0.08, 0.4, 0.9 use six-test cutoff 1/120, approximately 0.00833; only 0.004 rejects. Changed ten-test family has cutoff 0.005, with four outcomes missing; only 0.004 among supplied values rejects, and the other four decisions are unknown. Keep those tests in the denominator and disclose missing results. The scenarios are stipulated, not actual experiments.

  An adjusted p-value is min(1, m × p). Adjustment cannot validate invalid p-values, hide outcome-driven selection, prove non-rejected nulls or establish useful effect sizes. The error bound can be conservative and is not an observed error rate.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Four valid tests form one declared family with alpha 0.05. Their p-values are .01, .02, .2, .9. Use Bonferroni: divide family alpha by four and reject when p is at most that threshold. Compute the threshold and all four decisions.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **multiple comparisons evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **multiple comparisons**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Multiple comparisons**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Multiple comparisons**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Four valid tests form one declared family with alpha 0.05. Their p-values are .01, .02, .2, .9. Use Bonferroni: divide family alpha by four and reject when p is at most that threshold. Compute the threshold and all four decisions.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

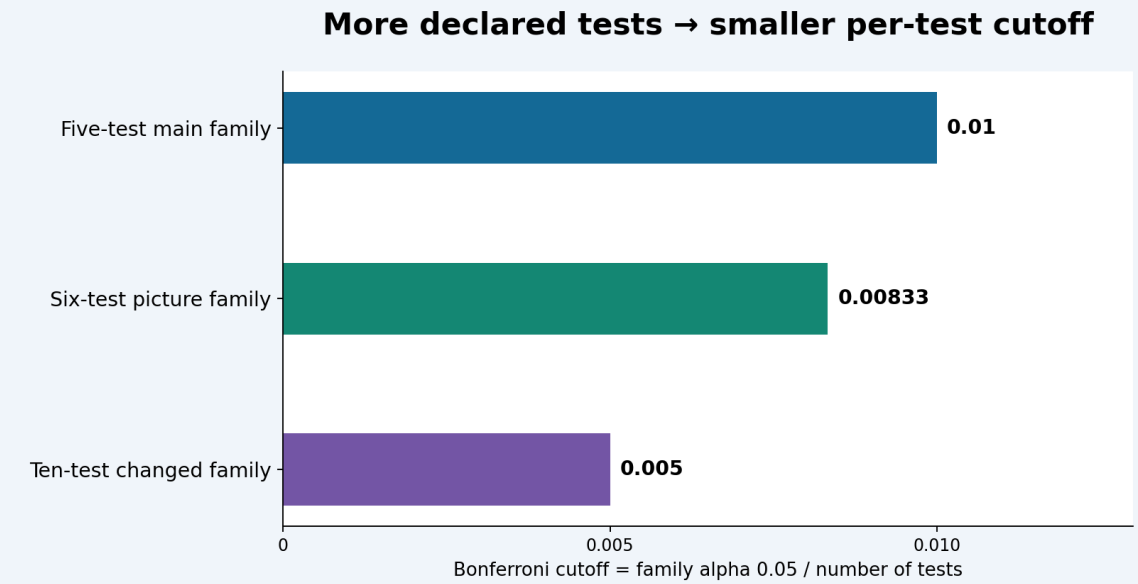
- Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

- Familywise alpha=0.05. Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

- Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



Reject a supplied valid test only when p ≤ its family cutoff, including equality.
Ten-test case: four p-values are missing; their decisions remain unknown.
These are decision boundaries, not effect sizes or measured false-positive rates.

AI ACADEMY / AI-7/15

Supplied p-values and six-test decisions

	Supplied p	At cutoff 0.00833
Test 1	0.004	Reject
Test 2	0.012	Do not reject
Test 3	0.02	Do not reject
Test 4	0.08	Do not reject
Test 5	0.4	Do not reject
Test 6	0.9	Do not reject

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/15

Multiple comparisons

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Count the planned tests

There are six tests in the declared family.

2. Compute corrected threshold

0.05/6 is about 0.00833 for each test.

3. Apply all six decisions

Only 0.004 is below 0.00833; the other five do not reject under this rule.

CHANGE ONE THING

The declared family has ten tests instead of six; keep the listed values unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.
- Which labelled step matches this detail? Only 0.004 is below 0.00833; the other five do not reject under this rule.
- 1. Apply all six decisions
- 2. Compute corrected threshold
- 3. Count the planned tests
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply all six decisions Only 0.004 is below 0.00833; the other five do not reject under this rule.

Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required.

A rejected null is not automatically a large or useful improvement.

Changed-case question and optional hint

- Changed-case practice: The declared family has ten tests instead of six; keep the listed values unchanged. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.

The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

Repair and retain the record

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Review Uncertainty intervals (ai-7/12). Goal: Explain the **mechanism** and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Review Experimental units and independence (ai-7/9). Goal: Explain the **mechanism** and terminology of experimental units and independence. Short practice: Count independent plants and photo views.

Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Multiple comparisons: the case

AI ACADEMY / AI-7/15 · 01 · LOOK AT THE FACTS

Multiple comparisons: the case

The situation

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

Supplied evidence

Familywise alpha=0.05. Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

Explain multiple comparisons using the records and boundaries of this fictional investigation.

Follow the labelled picture with the matching explanation. Open the vector diagram

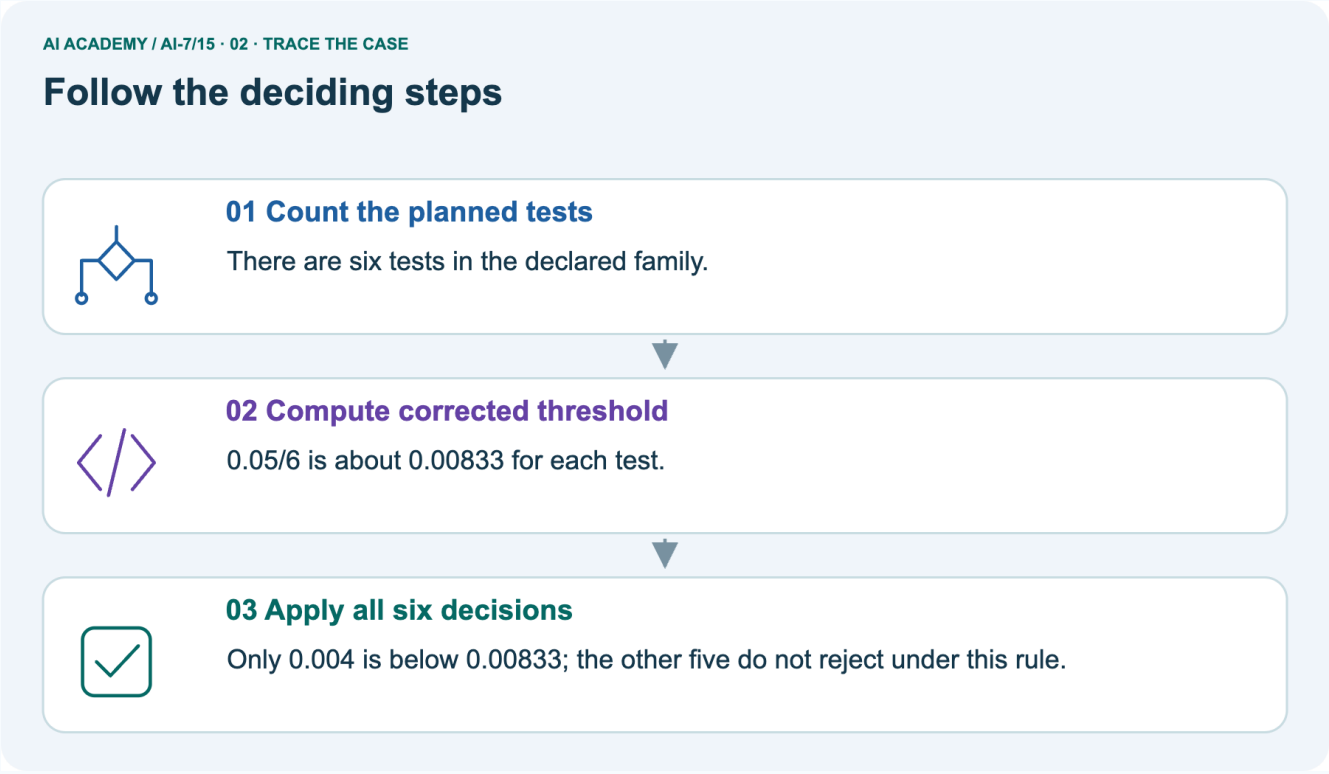
Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold. Supplied facts: Familywise alpha=0.05.

Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



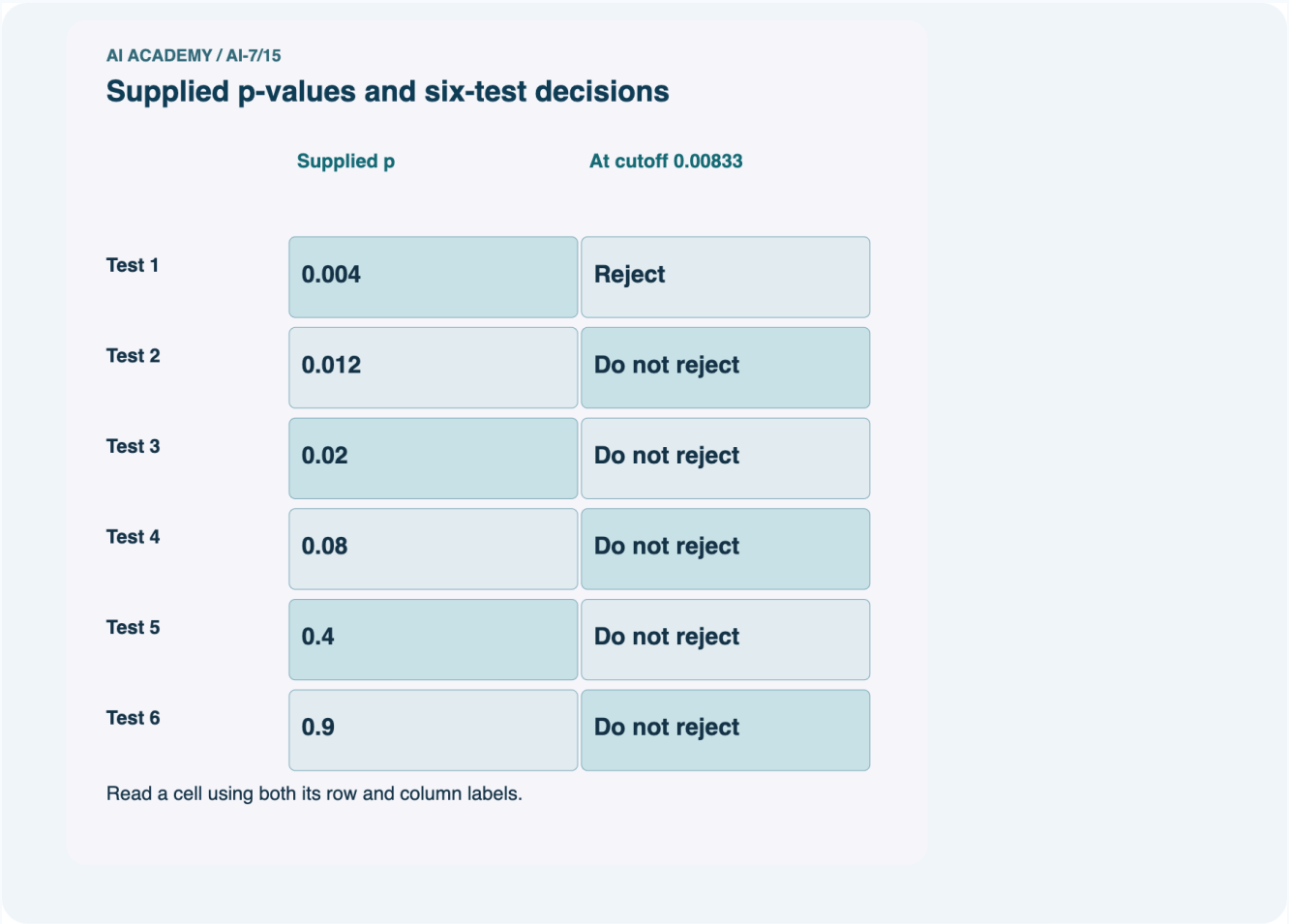
Follow the labelled picture with the matching explanation. Open the vector diagram

Count the planned tests: There are six tests in the declared family. → Compute corrected threshold: 0.05/6 is about 0.00833 for each test. → Apply all six decisions: Only 0.004 is below 0.00833; the other five do not reject under this rule.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Supplied p-values and six-test decisions



Follow the labelled picture with the matching explanation. Open the vector diagram

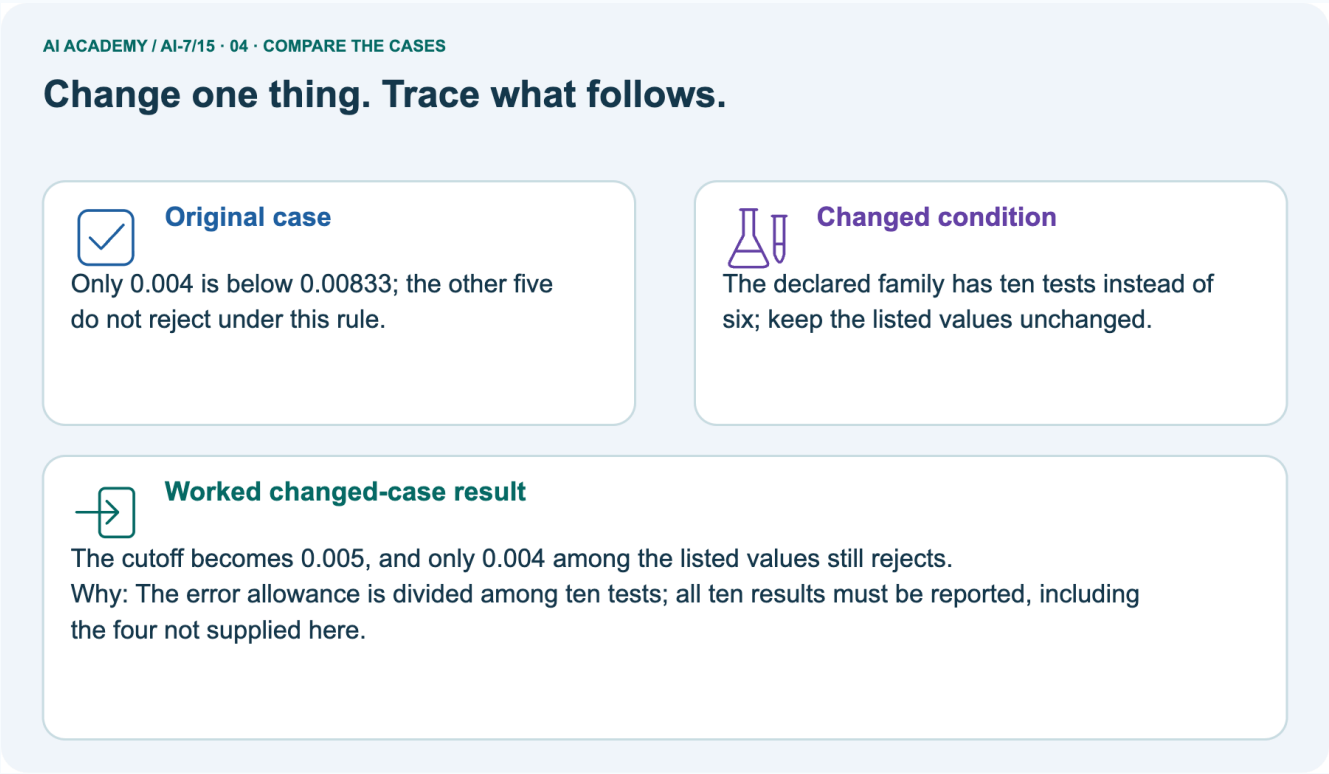
Supplied p-values and six-test decisions. Test 1: Supplied p = 0.004, At cutoff 0.00833 = Reject. Test 2: Supplied p = 0.012, At cutoff 0.00833 = Do not reject. Test 3: Supplied p = 0.02, At cutoff 0.00833 = Do not reject.

Test 4: Supplied p = 0.08, At cutoff 0.00833 = Do not reject. Test 5: Supplied p = 0.4, At cutoff 0.00833 = Do not reject. Test 6: Supplied p = 0.9, At cutoff 0.00833 = Do not reject.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

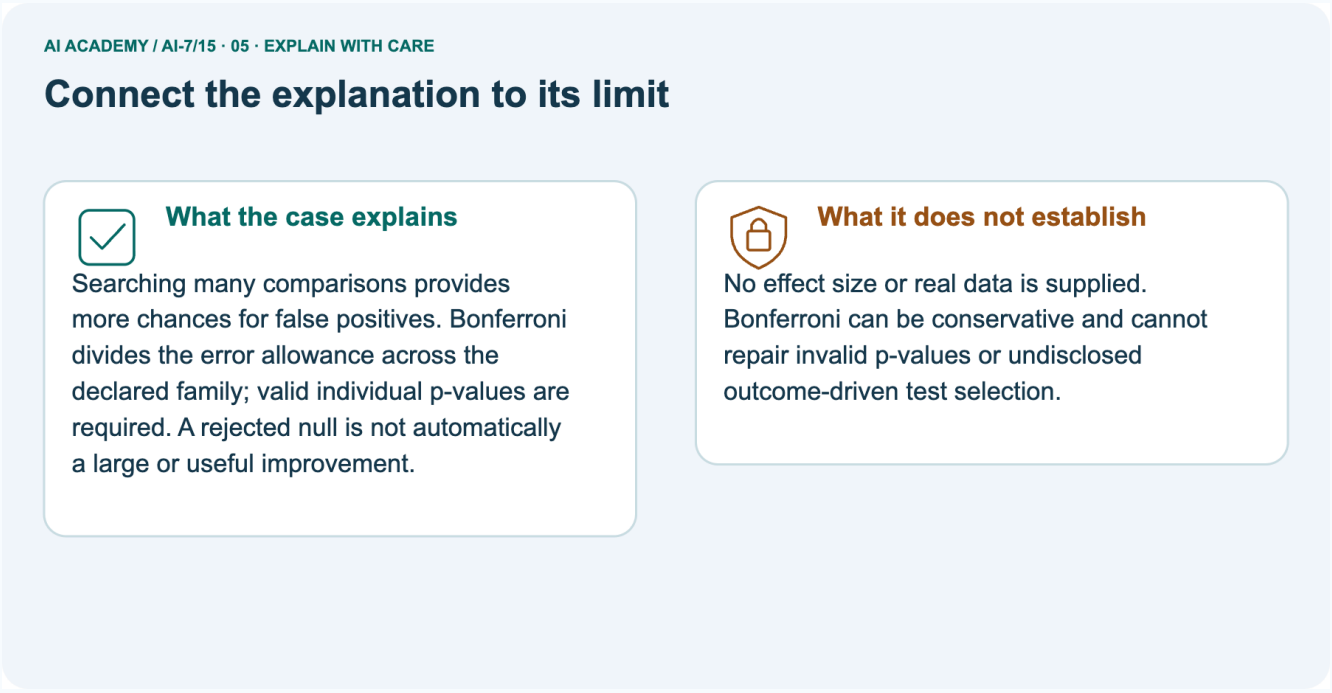
Original case: Only 0.004 is below 0.00833; the other five do not reject under this rule. Changed condition: The declared family has ten tests instead of six; keep the listed values unchanged.

Expected result: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects. Reason: The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required.

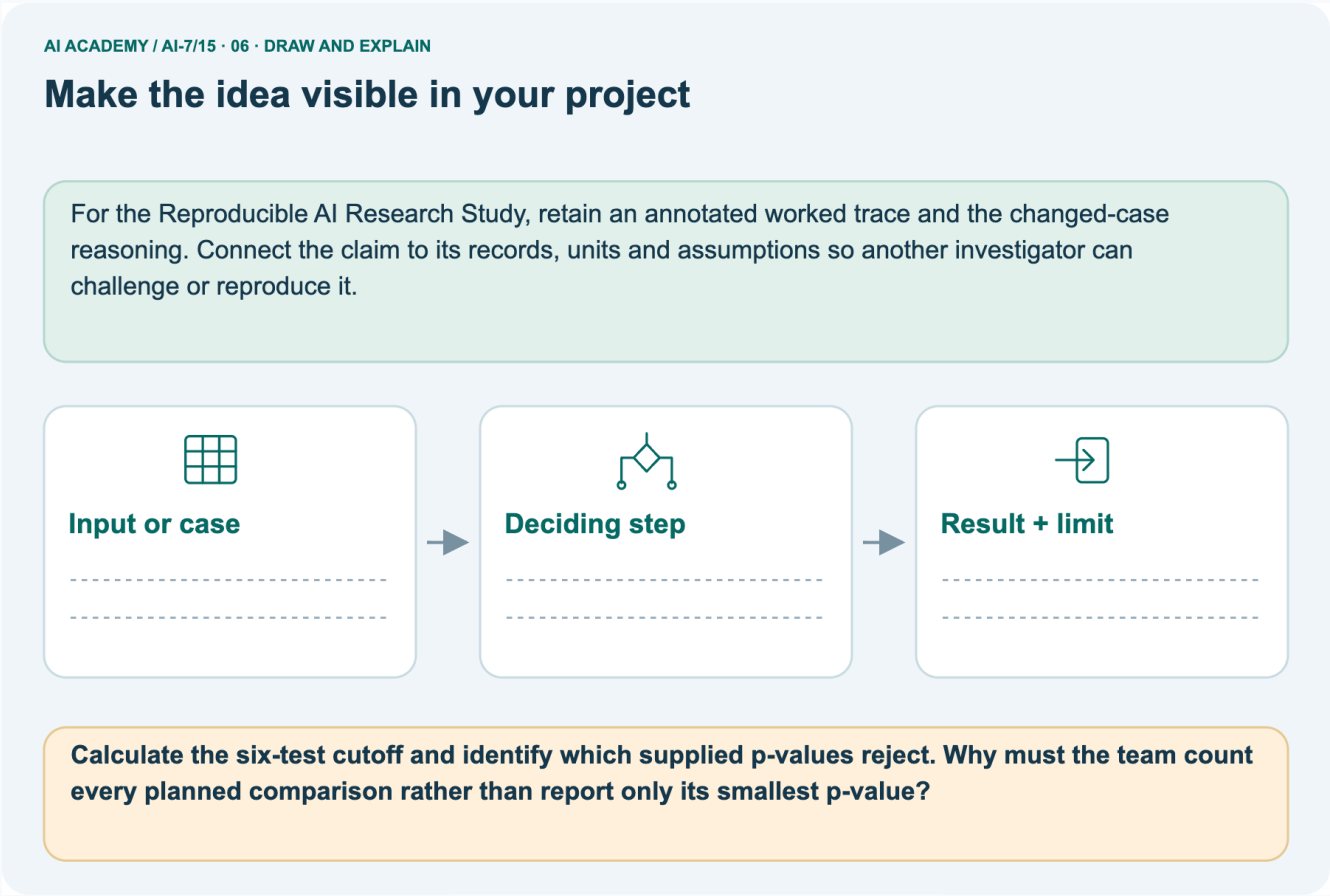
A rejected null is not automatically a large or useful improvement. Limit: No effect size or real data is supplied.

Bonferroni can be conservative and cannot repair invalid p-values or undisclosed outcome-driven test selection.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**
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▼ **Earlier wording, alternatives and picture transcripts**

01. Multiple comparisons: the case

02. Follow the deciding steps

03. Supplied p-values and six-test decisions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.